

Understanding TLC

No we are not promoting a travel channel, but this article gives you the dope on Triple Level Cell flash <http://goo.gl/1tYnY>

Hawley Creek to launch soon

Intel 313 series SSDs aka Hawley Creek are rumoured to launch alongside Ivy Bridge processors and series 7 chipsets

dled with a 2.5-inch to 3.5-inch adapter along with some screws.

Strontium Python series SSD's cover had information that was misleading. While the



Kingston HyperX 240GB SSD

front face advertises the SSD as having a SATA 3 interface, the rear side says that it supports SATA 2 interface only. Another discrepancy we found was the mention of SandForce SF1222 controller on the cover, whereas when we opened the drive, we found an SF 2281 controller just like the rest of the SSDs. So Strontium should pay some attention to the labelling. We found this drive to be surprisingly lighter than other drives. The type of MLC NAND chip used in this drive is hard to find out as the NAND chips have no manufacturer names.

Performance

Synthetic tests

We carried out three synthetic tests so as to have a broad overview of the SSDs across various benchmarks.

Crystal Disk Mark is a renowned utility known for testing drives and helps measure sequential and random read/write speeds. We kept the size at 100MB and ran 3 iterations of the test, so as to get a fair score. Under the All O Fill data we got scores which were sim-

ilar across all the SSDs. But you throw in random data, and not all SSDs perform equal. Since we were getting a difference in the default Random test data, we decided to evaluate the sequential and random read/write speeds for both the compressible (All O Fill) and incompressible (Random) test data.

SandForce controllers use compression technology while writing data for a better performance and longer SSD life. Random data is incompressible, hence you will get poor write speeds as

compared to compressible data. Crystal Disk Mark allows you to select between incompressible and compressible test data.

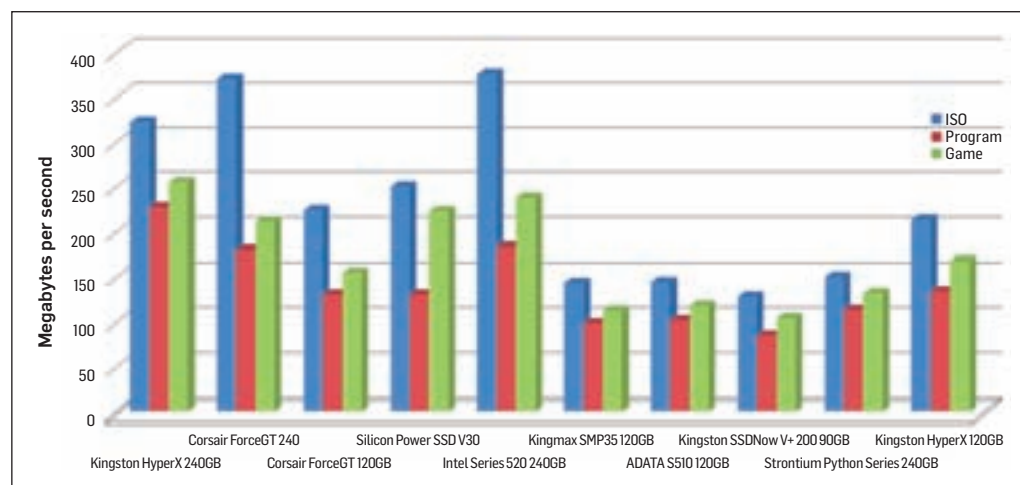
For O Fill test data the read and write scores were very close by as can be seen in the tables. This was true for all SSDs. But only the most powerful SSDs give similar scores when the test data is changed to Random. Kingston HyperX 240GB, Corsair Force GT 240 GB and Intel Series 520 SSDs gave much better sequential and random write speeds than their competitors. Notice the gap in read and write speeds for Corsair 120GB SSD and Silicon Power 120 GB SSD.

AS-SSD on the other hand only had incompressible test

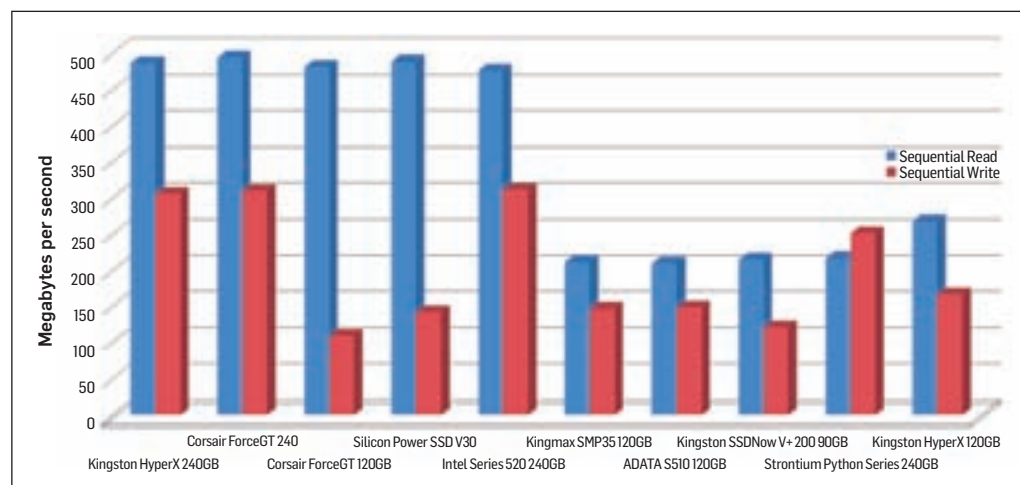
data. So while the sequential read speeds were great for most of the SSD with synchronous flash memory, the write speeds were equally bad. Only Kingston Hyper X 240 GB, Corsair 240 GB and Intel series 520 almost touched 300 MB/s in sequential writes.

We also carried out the copy test benchmark, which simulates a file copy scenario for three categories as shown below. Here again, the Intel series 520 and Corsair 240GB drives emerged with the best score.

Coming to PC Mark 7, we ran the secondary storage test which involves simulating importing pictures, video editing, starting applications



AS SSD Copy test



Crystal DiskMark Incompressible File Test (Seq Read / Write)